

Extraction of metamaterial constitutive parameters based on data-driven discontinuity detection

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Abstract: This paper presents an efficient data-driven method to extract the effective metamaterial parameters unambiguously. This method is based on detecting the discontinuity points in the real part of the refractive index and discerning the correct branch values at these discontinuities. The proposed method is numerically simple and does not require an infinite frequency integration. The performance of the proposed method is demonstrated by investigating thin and thick homogeneous and periodically structured metamaterial slabs. This method attains correct values of the branch index and, hence, correct values of the refractive index at the entire frequency range for thin and thick slabs independently from its imaginary part. It, notably, succeeds in overcoming the frequency and dimension limits and saturation observed in the Kramers–Kronig (K–K) method in thick homogeneous and multi-cell periodically structured slabs. The proposed method also gives accurate values for the metamaterial parameters at resonance and at the negative parameter region.

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1. Introduction

Metamaterials are artificially engineered composite structures that have properties not typically attainable by natural materials, such as a negative refractive index [1]. The macroscopic characterization of complex synthesized electromagnetic materials relies on relationships that arise from field averaging or effective medium homogenization. The defined effective electric permittivity ϵ_{eff} and effective magnetic permeability μ_{eff} encapsulate the specific local details of the composite medium [2]. ϵ_{eff} describes the response of metamaterials to the electric field, whereas μ_{eff} describes the response of metamaterials to the magnetic field [1]. Retrieval techniques to obtain the electromagnetic characteristics of metamaterials have received notable efforts, as presented in [2–12]. The retrieval techniques are, essentially, based on the plane-wave scattering parameters from a metamaterial slab in free space [4,11]. Expressions of the transmission and reflection S-parameters as functions of the material complex refractive index and wave impedance are obtained from the relationship between the scattering and the transmission matrices. The retrieval of the metamaterial parameters starts from deducing the values of the effective refractive index n_{eff} and the effective impedance z_{eff} from the S-parameters. ϵ_{eff} and μ_{eff} are then extracted from the basic relations: $\epsilon_{eff} = n_{eff}/z_{eff}$ and $\mu_{eff} = n_{eff}z_{eff}$. However, this method is liable to obtain a refractive index with multi-branching ambiguity due to the logarithmic nature of its real part function [3,4,13]. This ambiguity is manifested when the thickness of the metamaterial slab is large or the frequency region investigated is high. A nonlocal constitutive relations are demonstrated in [14] for metamaterial with strong spatial dispersion instead of local constitutive relations that are valid for weak spatial dispersion. The electromagnetic response of metamaterials is usually developed from material description involving bulk or tensorial representation [15]. This description, however, is only approximate, as spatial dispersion is always present to some degree in metamaterials [9,16]. Determining the correct value of the real refractive index is challenging, especially in the case of dispersive materials [11]. Chen et al. [3]

used a Taylor expansion approach considering the fact that the refractive index n_{eff} is a continuous function of frequency to select the proper branch. This study was performed for frequency regions below and above the resonance band excluding the resonance region itself. However, applying this approach to the resonance region gives wrong branch index for all frequency points after resonance. The Kramers–Kronig (K–K) [17] integral relation between the real and the imaginary parts of the refractive index was introduced in [4,18] to remove the ambiguity of the real refractive index. In other words, the K–K-based methods rely on the relationship between the imaginary and the real parts of the refractive index using the fact that the imaginary part is not affected by the branching problem. The integral limits of the K–K relation are 0 and ∞ . Therefore, the values of the S-parameters must be determined for the entire frequency range to get the correct solution. Practically, the integral must be truncated while keeping the range as large as possible. In addition, this method is saturated at high κd values, which limits its performance at high frequencies, especially for thick structures. Some improved approaches use phase correction techniques of the S-parameters or composed functions of the refractive index near the resonance region to avoid the heavy computation of the Kramers-Kronig approach [19–21]. The unwrapping function used for phase correction needs to determine the starting frequency that has zero branching [20] which is typically set to zero. However, when the desirable frequency band extends to high frequencies, the phase unwrapping technique becomes computationally inefficient due to computations at regions out of the concerned frequency band [18]. In addition, phase of the refractive index is known to be sensitive to small ripples in the S-parameters measurements at low frequencies especially for multiple layer metamaterials [4]. The resulted complex refractive index will suffer from inaccuracies in the phase term. Different phase unwrapping approaches have been proposed to address these problems as in [19–23]. Nicolson–Ross–Weir (NRW) relations were used with the K–K integrals to extract the unambiguous parameters of metamaterial slabs [12,13]. However, the limitations in the high frequency ranges for thick material slabs still exist. The K–K method is still the most effective approach to solve the branching ambiguity problem in the values of effective metamaterial parameters.

This paper develops an efficient retrieval method based on S-parameters to accurately extract unambiguous ε_{eff} and μ_{eff} of metamaterials. This method is based on detecting the discontinuity points in the real part of the refractive index. The results of the proposed method are compared to those of the K–K method to demonstrate the performance of the proposed method. Two examples are used in this study: uniform homogenous slabs and split-ring resonator (SRR) structure metamaterial slabs.

2. Retrieval method formulation

In the proposed data-driven (D-D) method, the effective parameters of a metamaterial slab are retrieved from the free space reflection and transmission coefficients. The complex S-parameters, at N distinct frequency points, can be obtained by measurements or from full-wave electromagnetic simulators.

For normal incident plane waves on a homogeneous metamaterial slab, as depicted in Fig. 1, the relationships between the S-parameters with the complex wave impedance and the complex refractive index [3,4] are given by

$$S_{11} = \frac{\left(\frac{z_{eff}-1}{z_{eff}+1}\right) (1 - e^{i2n_{eff}\kappa d})}{1 - \left(\frac{z_{eff}-1}{z_{eff}+1}\right)^2 e^{i2n_{eff}\kappa d}} \quad (1)$$

$$S_{21} = \frac{\left(1 - \left(\frac{z_{eff}-1}{z_{eff}+1}\right)^2\right) e^{in_{eff}\kappa d}}{1 - \left(\frac{z_{eff}-1}{z_{eff}+1}\right)^2 e^{i2n_{eff}\kappa d}} \quad (2)$$

where d is the metamaterial slab thickness, κ is the free-space wavenumber, z_{eff} is the complex effective wave impedance ($z_{eff} = R + iX$), and n_{eff} is the complex effective refractive index ($n_{eff} = n' + in''$). z_{eff} and n_{eff} are obtained by inverting Eqs. (1) and (2) [3,11], yielding

$$z_{eff} = \pm \sqrt{\frac{(1 + S_{11})^2 - S_{21}^2}{(1 - S_{11})^2 - S_{21}^2}} \quad (3)$$

$$n_{eff} = \frac{-i}{\kappa d} \ln \left(\frac{S_{11}}{1 - S_{21} \left(\frac{z_{eff}-1}{z_{eff}+1} \right)} \right). \quad (4)$$

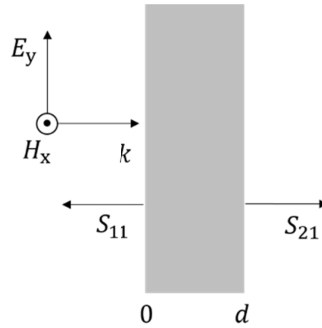


Fig. 1. Metamaterial slab with normal incident plane wave.

For passive metamaterials, the real value of the complex wave impedance and the imaginary value of the complex refractive index must be greater or equal to zero; $R \geq 0$, $n'' \geq 0$. Therefore, the sign of z_{eff} must be determined according to those conditions.

The real and the imaginary parts of the complex refractive index are given, explicitly, by [3,4]

$$n' = \frac{1}{\kappa d} \operatorname{Im} \left(\ln \left(\frac{S_{11}}{1 - S_{21} \left(\frac{z-1}{z+1} \right)} \right) \right) + \frac{2m\pi}{\kappa d} \quad (5)$$

$$n'' = \frac{-1}{\kappa d} \operatorname{Re} \left(\ln \left(\frac{S_{11}}{1 - S_{21} \left(\frac{z-1}{z+1} \right)} \right) \right) \quad (6)$$

where m is the branch index of the multivalued logarithmic function [4]. The resulting branch uncertainty in the real part of the complex refractive index leads to ambiguities in the expressions of the effective permittivity and effective permeability of the metamaterial. The complex permittivity and complex permeability are related to the complex refractive index and the wave impedance by the following expressions [11]:

$$\varepsilon_{eff} = \frac{n_{eff}}{z_{eff}} \quad (7)$$

$$\mu_{eff} = n_{eff} z_{eff}. \quad (8)$$

For a small metamaterial thickness (d), m is set to zero, as presented in [11]. For a large d , selection of the correct value of m becomes uncertain and may lead to flawed parameter extraction, as outlined in [4]. To overcome this problem, the authors of [4,18] used an extraction method

based on the Kramers–Kronig relationship. Knowing the imaginary part of the refractive index, the real part of the refractive index can be determined according to the K–K relationship [4] from

$$n'^{K-K}(f) = 1 + \frac{2}{\pi} \mathcal{P} \int_0^{\infty} \frac{f' n''(f')}{f'^2 - f^2} df' \quad (9)$$

where $\mathcal{P}$ is the principal value of the improper integral [4,15], and according to [4], the correct branch index is calculated as

$$m(f) = \text{Round} \left[(n'^{K-K}(f) - n'(f)) \frac{\kappa d}{2\pi} \right]. \quad (10)$$

However, there is a limitation for this method due to the truncation in the upper limit of the K–K integral [4]. The requirement to select the proper values of the branch index at high frequencies is to extend the integral band to the largest possible value, which is practically unmanageable. In addition, even with the frequency interval extension, it is found that this K–K integral-based method is saturated at a maximum κd value of roughly 3.17. In the presented work, we developed a novel method to select the correct value of the branch index in n' independently of the n'' , based on a simple numerical algorithm not involving the cumbersome K–K integration relation. The proposed method is independent of the frequency bands and gives accurate values of the branch index for any frequency region. The proposed algorithm is data driven; it deals with the measured S parameters numerical data to detect and correct for the branching discontinuity in the refractive index regardless of the orientation of the measured data.

3. Data-driven discontinuity detection method

The proposed method is established based on the fact that the behaviors of ε_{eff} and μ_{eff} of any material are continuous with frequency [3,4,11,24]. The continuity of ε_{eff} and μ_{eff} implies the continuity of the real part of the complex refractive index. This method relies on preserving the continuity in the real part of the refractive index and, hence, observing and correcting the multivalued branching of n' with frequency. The break points in n' are detected by observing its frequency rate of change using the first-order differential equation given by

$$\mathcal{D}(f) = \frac{dn(f)}{df}. \quad (11)$$

Discretization of Eq. (11) yields

$$\mathcal{D}(f_i) = \frac{n'(f_i) - n'(f_{i-1})}{f_i - f_{i-1}}. \quad (12)$$

If $\mathcal{D}(f_i)$ satisfies the condition, $|\mathcal{D}(f_i)| \neq |q(f_i)|$, where q is the branching singular value, explained later, the branch index, m , does not change its value and n' does not acquire a new branch. However, if $|\mathcal{D}(f_i)| = |q(f_i)|$, branching occurs and m should acquire another value to preserve the frequency continuity of ε_{eff} , μ_{eff} , and, hence, $n'(f_i)$. The branching change of m is calculated according to the following equation:

$$m_i = \begin{cases} \text{Round} \left[(n'(f_{i-1}) - n'(f_i)) \frac{\kappa d}{2\pi} \right] & \text{for } |\mathcal{D}(f_i)| = |q(f_i)| \\ 0 & \text{Otherwise} \end{cases} \quad (13)$$

where $i = 1, 2, \dots, N-1$, $m_0 = 0$ is the low frequency branching number, and N is the number of distinct frequency points.

Using the branching condition, $|\mathcal{D}(f_i)| = |q(f_i)|$, ensures the small jumps at resonance frequency regions are not interpreted as a branching problem. The total branch index is the accumulative summation of the branching changes given by

$$m = \sum_{i=0}^{N-1} m_i. \quad (14)$$

It is noted that at the break points, the n' frequency curve, for $m = 0$, has an abrupt behavior by inverting the sign of the n' value from $\pm x$ to $\mp x$, where x is positive [4]. Consequently, the amount of change at the branching discontinuity between two successive points is about $2x$. Therefore, the branching singular value q can be given by the following equation:

$$q(f_i) = \frac{2n'(f_{i-1})}{(f_i - f_{i-1})}. \quad (15)$$

4. Parameter extraction of a uniform homogenous slab

In this section, a uniform homogenous metamaterial slab is used to demonstrate the applicability of the proposed method compared to the K-K method and to estimate the accuracy of the extracted effective material parameters. The behaviors of the z_{eff} and n_{eff} indices of many synthesized metamaterials (especially single-unit-cell and two-unit-cell metamaterial slabs [25]) are similar to that of the effective uniform homogenous slab that is modeled with the Drude and Lorentz models [4,25]. The left handed metamaterials can be described by Drude-Lorentz frequency model of their effective permittivity and permeability, respectively, when the plasma frequency is greater than the magnetic resonant frequency [26,27]. Three different thicknesses of uniform homogenous material are employed (i.e., 40, 200, and 400 nm) to demonstrate the applicability of the proposed method.

The effective electric permittivity of uniform homogenous material is given by the Drude model as

$$\epsilon_{eff} = \epsilon_{\infty} - \frac{\omega_p^2}{\omega^2 + i\gamma_c\omega}. \quad (16)$$

However, the magnetic permeability is modeled by the Lorentz model as

$$\mu_{eff} = \mu_{\infty} - \frac{(\mu_s - \mu_{\infty})\omega_0^2}{\omega^2 + i\delta\omega - \omega_0^2}. \quad (17)$$

The effective electric permittivity and the magnetic permeability in the above models are functions of frequency ω and independent of the thickness d . ϵ_{∞} and μ_{∞} , respectively, are the electric permittivity and magnetic permeability at high frequencies, ω_p is the Drude plasma frequency, and ω_0 is the magnetic resonant frequency. γ_c , δ , and μ_s are the collision frequency, the magnetic damping factor, and the static magnetic permeability, respectively. The parameters of the investigated homogeneous metamaterial slabs are given in Table 1.

Figure 2 shows ϵ_{eff} and μ_{eff} of the Drude and Lorentz models according to Eqs. (16) and (17). The frequency range is from 0 to 1.5 PHz, where PHz is 10^{15} Hz.

The reflection and transmission coefficients of a uniform plane wave normally incident on those uniform homogenous slabs ($d = 40$ nm, $d = 200$ nm and $d = 400$ nm) are computed and presented in Figs. 3, 4, and 5 respectively. They are determined according to Eqs. (1) and (2) considering the passive material conditions. It is obvious that discontinuities appear in the phase of the transmission coefficient S_{21} for the cases of $d = 200$ nm and $d = 400$ nm. Discontinuities are more prevalent in thicker slabs.

Using the S-parameter retrieval method, z_{eff} and n_{eff} are extracted, as shown in Fig. 6. Figures 6(a) and (c) show that the real part of z_{eff} and the imaginary part of n_{eff} satisfy the passive

Table 1. Drude and Lorentz Parameters of Investigated Slabs.

ϵ_{∞}	1.8	
μ_{∞}	2.2	
ω_p	$2\pi \times 0.8 \times 10^{15}$	rad/s
ω_0	$2\pi \times 0.4 \times 10^{15}$	rad/s
γ_c	80×10^{15}	s^{-1}
δ	0.05×10^{15}	s^{-1}
μ_s	2.6	

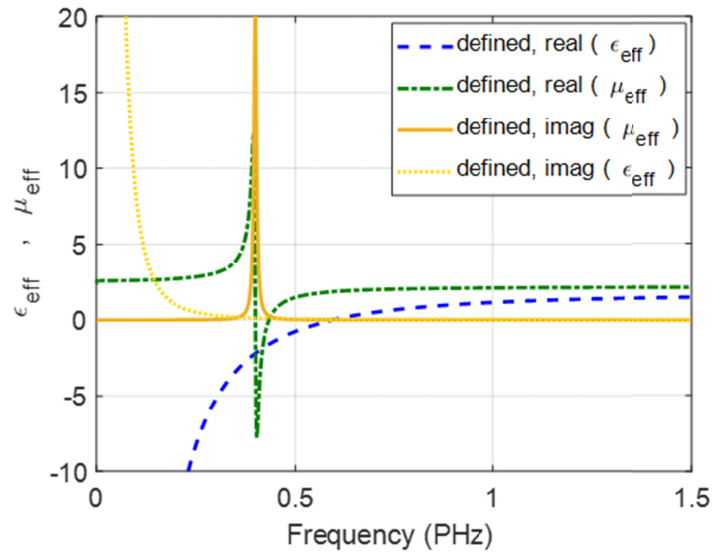
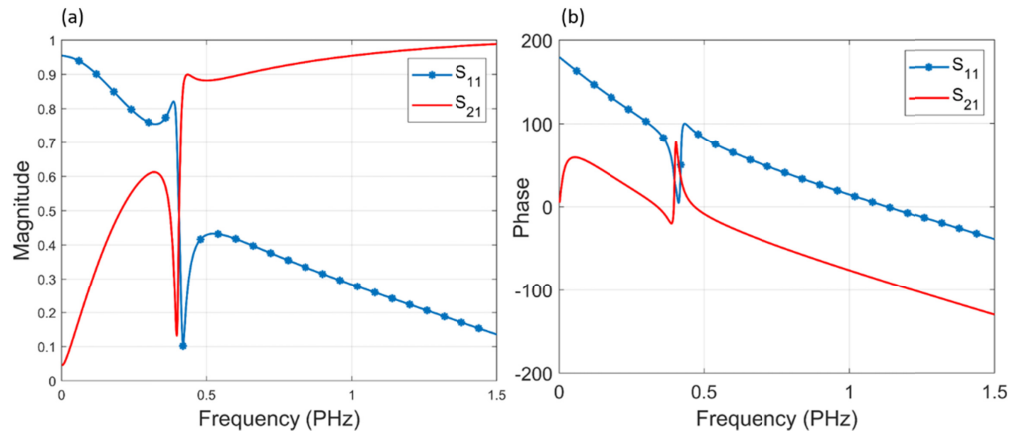


Fig. 2. Permittivity of Drude model and permeability of Lorentz model.

Fig. 3. (a) Magnitude and (b) phase of reflection and transmission coefficients for $d = 40 \text{ nm}$.

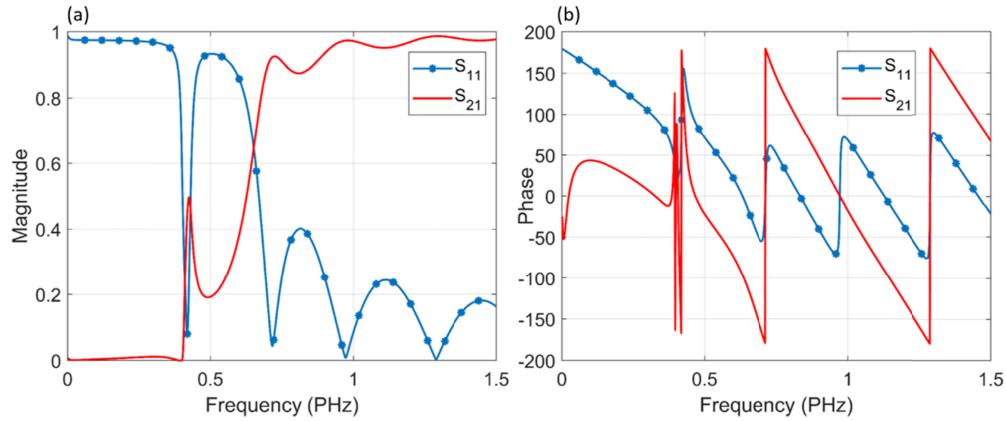


Fig. 4. (a) Magnitude and (b) phase of reflection coefficients for $d = 200$ nm.

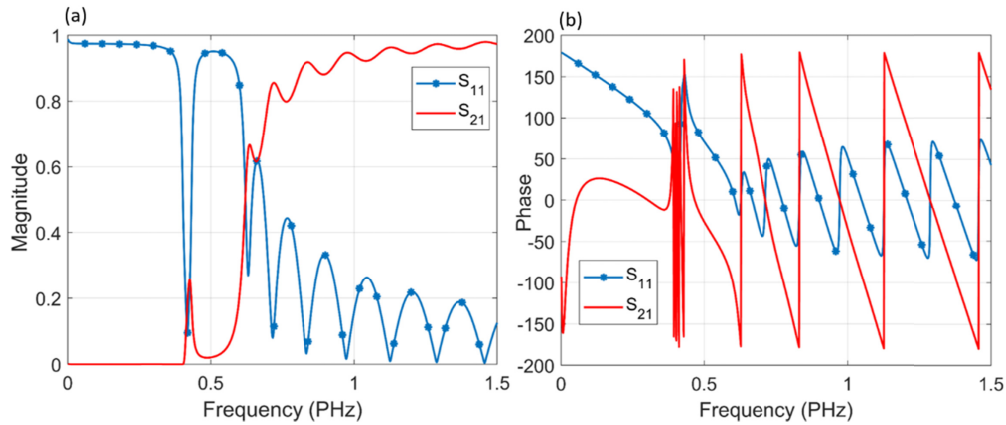


Fig. 5. (a) Magnitude and (b) phase of reflection coefficients for $d = 400$ nm.

material conditions. Figure 6(b) shows the imaginary part of z_{eff} . It is also illustrated in Fig. 6(d) that there is no branching in the real refractive index for the $d = 40$ nm slab. However, there is multiple branching in the real refractive index for the slabs of $d = 200$ nm and $d = 400$ nm at higher frequencies.

Both the K–K and the D–D methods are used to find the real part of the refractive index using its corresponding branch index. Figures 7, 8, and 9 show the real refractive index and its branch indices for $d = 40$ nm, $d = 200$ nm, and $d = 400$ nm. For $d = 40$ nm, the real refractive index has no discontinuities and the extracted permittivity and permeability are similar to the defined values. For the slab with $d = 200$ nm, the K–K method corrects the branching problem at low frequencies $\kappa d < 3.07$. The extension of the upper frequency limit of the K–K integration relation can improve the extraction performance for the frequency region $3.07 < \kappa d < 3.17$. At the point ($\kappa d \approx 3.17$, $f = 0.756$ PHz for the 200 nm slab), the extension of the frequency limit of the K–K method saturates and does not give correct results. The proposed method successfully extracts the correct branch index for low and high frequencies independent of the frequency region and slab width. For $d = 400$ nm, the K–K method fails to overcome the branching at the break point $\kappa d \approx 3.17$, $f = 0.377$ PHz just below the resonance region with the negative real

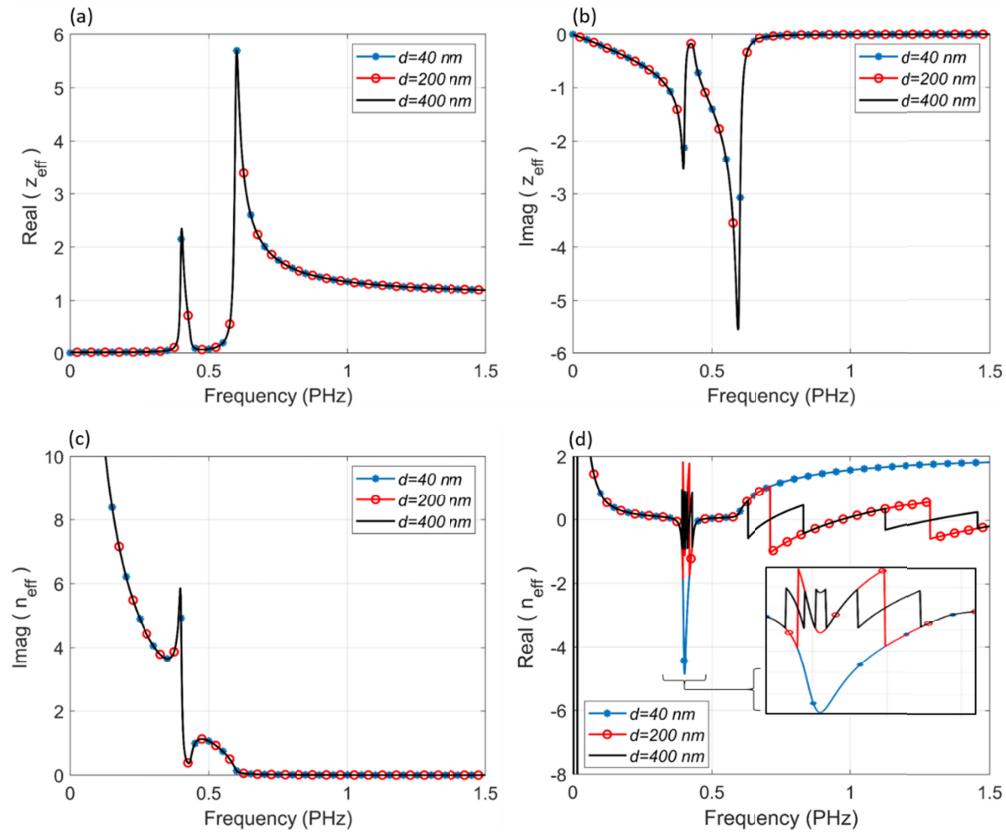


Fig. 6. Wave impedance and refractive index of homogeneous slabs ($d = 40$ nm, $d = 200$ nm, and $d = 400$ nm): (a) The real part of complex wave impedance, (b) imaginary part of complex wave impedance, (c) imaginary part of refractive index and (d) real part of refractive index.

n_{eff} . The presented scheme successfully extracts the correct values of the negative n' for all slab thicknesses and at the entire frequency band.

Figures 10, 11, and 12 show the extracted effective permittivity and permeability for different slab thicknesses compared with the predefined parameters. When $d = 40$ nm, both the K-K and the D-D methods give the correct effective permittivity and permeability as the predefined parameters. For the $d = 200$ nm slab with a resonance region at $\kappa d = 1.689$, both methods successfully yield the true values of ϵ_{eff} and μ_{eff} at the lower frequencies ($\kappa d < 3.17$, $f < 0.756$ PHz). However, the K-K method gives an incorrect real refractive index for $\kappa d > 3.17$, whereas the proposed method gives a correct n' for the entire frequency band, as shown in Fig. 11(b). For the $d = 400$ nm slab, with a resonance region at $\kappa d = 3.378$, the K-K method produces incorrect values of ϵ_{eff} and μ_{eff} at higher frequencies $\kappa d > 3.17$ regardless of the additional frequency limit extension, while the proposed method directly extracts the correct values of ϵ_{eff} and μ_{eff} , as illustrated in Fig. 12(b).

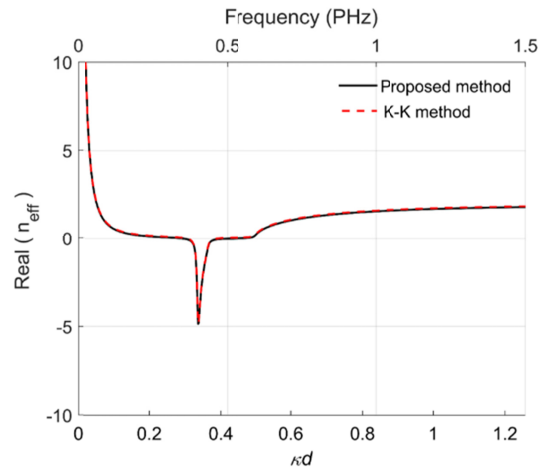


Fig. 7. Real part of the refractive index of the homogenous slab ($d = 40$ nm).

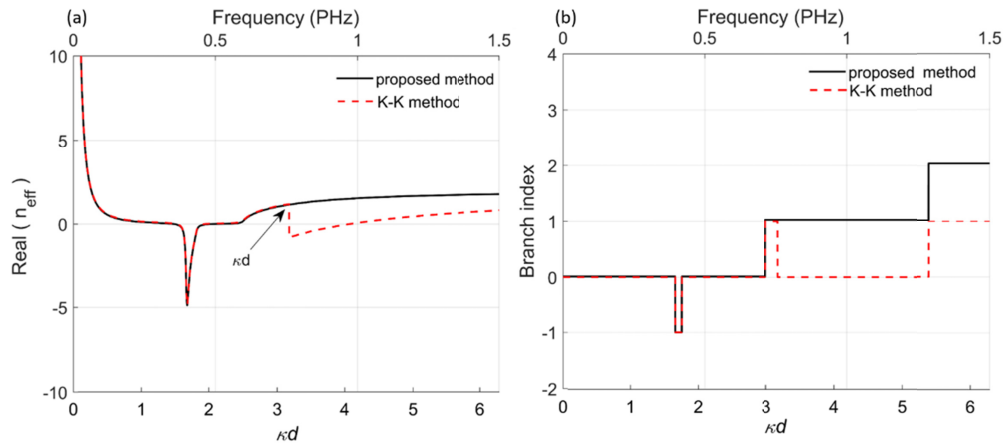


Fig. 8. (a) Real part of the refractive index of the homogenous slab ($d = 400$ nm). (b) Branch index of the homogenous slab ($d = 200$ nm).

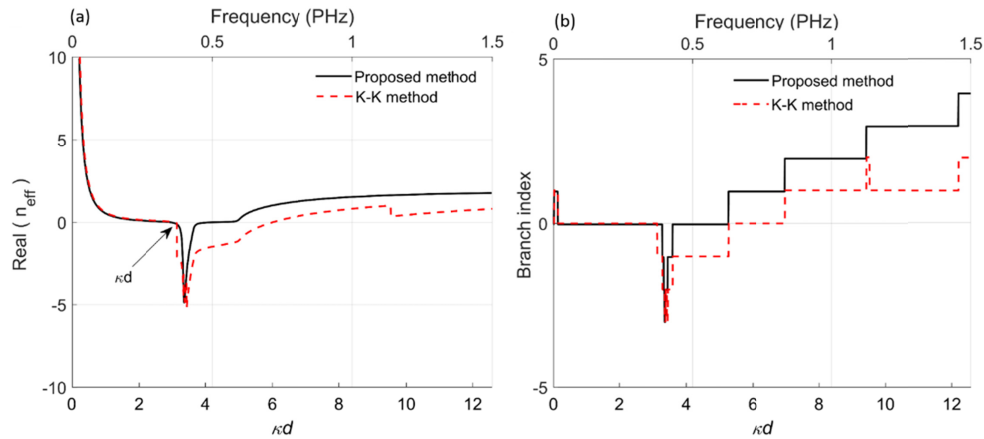


Fig. 9. (a) Real part of the refractive index of the homogenous slab ($d = 400$ nm). (b) Branch index of the homogenous slab ($d = 400$ nm).

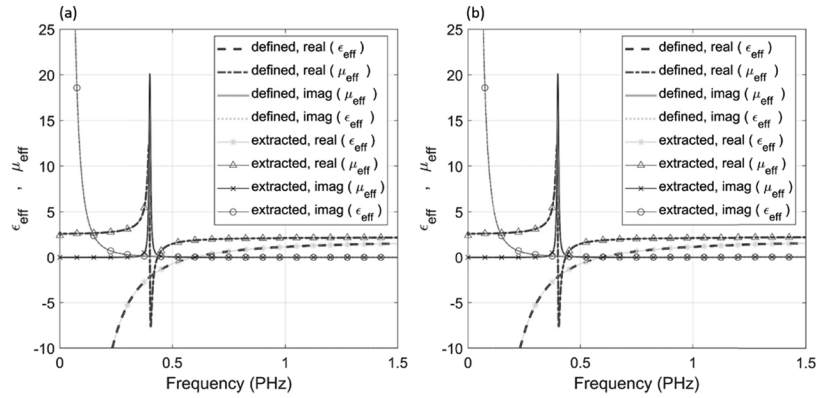


Fig. 10. The extracted and defined effective permittivity and permeability parameters of homogenous slab for $d = 40 \text{ nm}$ using (a) K-K method and (b) D-D method.

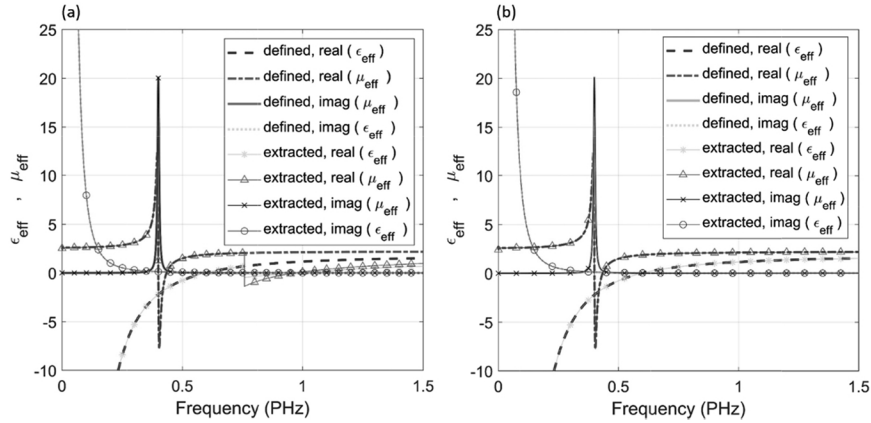


Fig. 11. The extracted and defined effective permittivity and permeability parameters of homogenous slab for $d = 200 \text{ nm}$ using (a) K-K method and (b) D-D method.

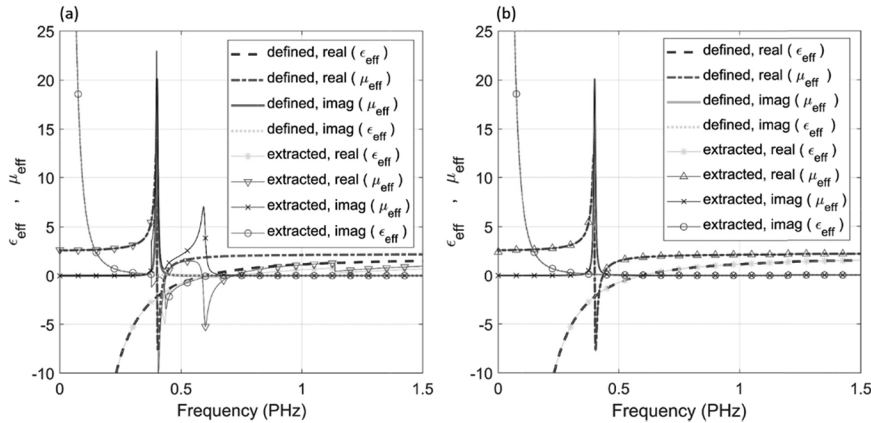


Fig. 12. The extracted and defined effective permittivity and permeability parameters of homogenous slab for $d = 400 \text{ nm}$ using (a) K-K method and (b) D-D method.

5. Parameter extraction of a metamaterial slab

A well-studied metamaterial periodic structure that yields negative permittivity and negative permeability presented in [2–4,11] consists of metallic rods and square SRR-rod separated by a dielectric slab. The unit cell geometry is shown in Fig. 13. The unit cell dimension is set to 2.5 mm. The width of the dielectric slab ($\epsilon = 4.4$, $\tan\delta = 0.02$) is 0.25 mm. The width of the metallic rod and the conductor of the ring are 0.2 mm and 0.14 mm, respectively, both with a thickness of 17 μm . The ring gap is set to 0.3 mm. The lengths of the inner and outer square rings are 1.5 mm and 2.2 mm, respectively. The direction of wave propagation is in the $-z$ direction.

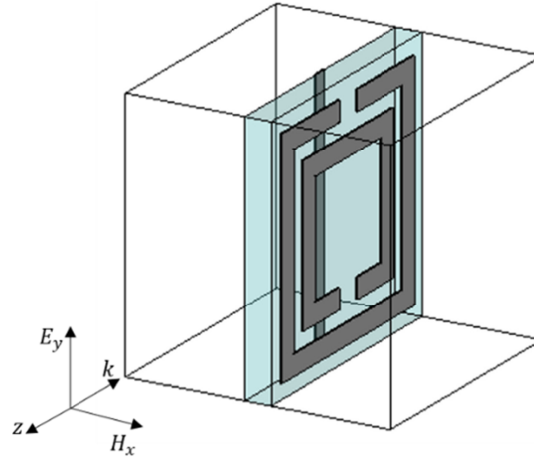


Fig. 13. Unit cell structure of metamaterial geometry consisting of metallic rods and SRRs separated by dielectric.

The S-parameters for a normal plane wave incident on the metamaterial slab are computed using a full-wave electromagnetic simulator. Then, the complex refractive index n_{eff} and the complex normal wave impedance z_{eff} are computed accordingly.

The real part of the wave impedance and the imaginary part of the refractive index satisfy the passive material conditions. The real part of the refractive index n' coincides with the result from the proposed method for $f < 29.8 \text{ GHz}$ ($\kappa d < 1.56$), as illustrated in Fig. 14. Both the K–K method and the proposed method successfully correct the branching between $23.5 \text{ GHz} < f < 25.4 \text{ GHz}$. However, a branching discontinuity occurs at 29.8 GHz for the K–K method. This branching discontinuity can be resolved only by substantial extension of the frequency band limit of the integral.

The branch correction outcome of the K–K method and the D–D method (proposed method) for a structure with a one-unit cell thickness (in the z direction) is shown in Fig. 15. The first branch (at 23.5GHz) and the second (at 25.4 GHz) are solved correctly in both methods. However, incorrect branch detection occurs at 29.8 GHz for the K–K method, which yields an incorrect value of n' for $f > 29.8 \text{ GHz}$.

Figure 16 shows the real refractive index of two-unit cells ($d = 5 \text{ mm}$) for the K–K method and the proposed method. In the region between $9.24 \text{ GHz} < f < 9.99 \text{ GHz}$ ($0.967 < \kappa d < 1.04$), the K–K method gives an incorrect n' . This problem appears in both a homogenous thick slab ($d = 400 \text{ nm}$) and a two-cell SRR-rod metamaterial slab. Extending the integral upper frequency limit up to 30 GHz in the K–K method corrects the branching issue that occurs at 25.8 GHz ($\kappa d = 2.7$). The method proposed in this paper directly extracts the correct n' for the entire band, as indicated by the black line with circle markers in the figure.

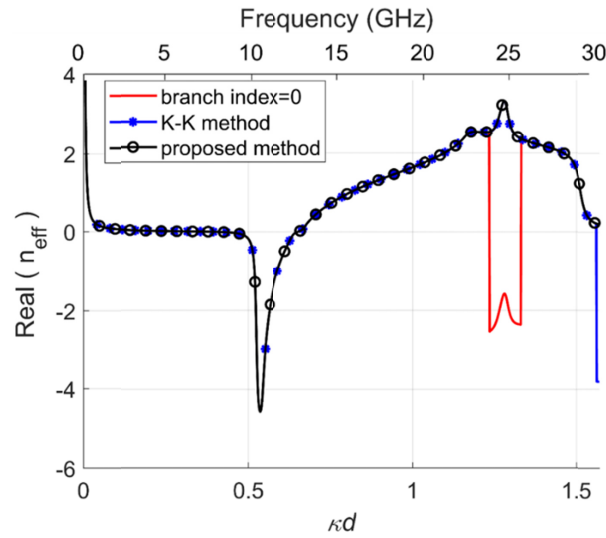


Fig. 14. Refractive index of one-unit cell SRR-rod (in z direction).

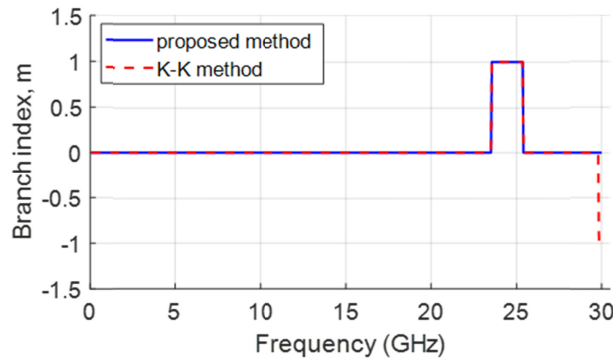


Fig. 15. Branch index of one-unit cell SRR-rod.

Figure 17 depicts that for the two-unit cell metamaterial slab, the K–K method as compared to the proposed method shows incorrect branching at frequency bands ($9.24 \text{ GHz} < f < 9.99 \text{ GHz}$) and for band $f > 25.8 \text{ GHz}$.

Figure 18 shows the corresponding results for the effective permittivity and permeability of a one-unit cell ($d = 2.5 \text{ mm}$). For $f > 42.8 \text{ GHz}$, the K–K method has a clear branching discontinuity in the effective permittivity and permeability. For the two-unit cell ($d = 5 \text{ mm}$) metamaterial slab, the K–K method gives discontinues values of the effective permittivity and the effective permeability for $f > 25.8 \text{ GHz}$, as shown in Fig. 19. Increasing the slab thickness reduces the performance frequency range of the K–K method. On the contrary, the D–D method yields continuous values of the effective permittivity and permeability regardless of the slab thickness and the frequency region.

The proposed technique succeeded to overcome the branching discontinuity problem efficiently and extended the applicability of the simple linear polarization normal incidence extraction procedures. Further, the presented algorithm can be applied to correctly extract the directional constitutive parameter from their measured numerical data for a tensorial representation of permittivity and permeability.

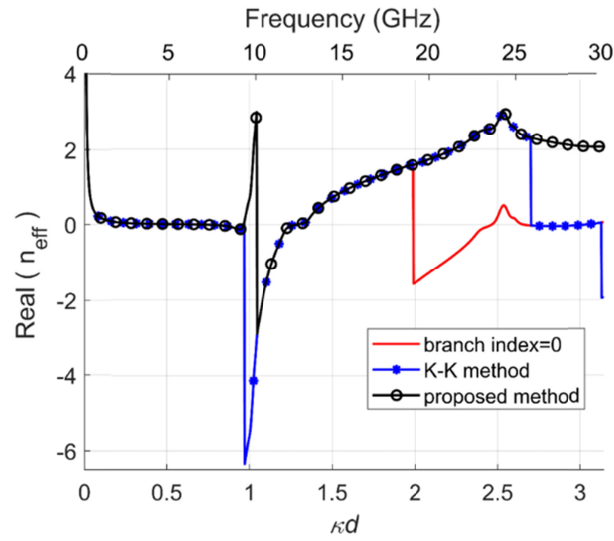


Fig. 16. Refractive index of two-unit cell SRR-rod.

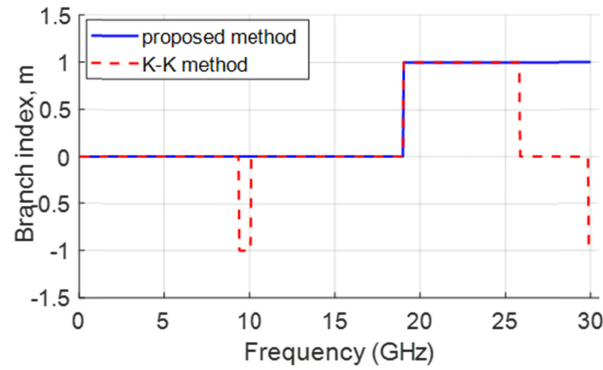


Fig. 17. Branch index of two-unit cell SRR-rod.

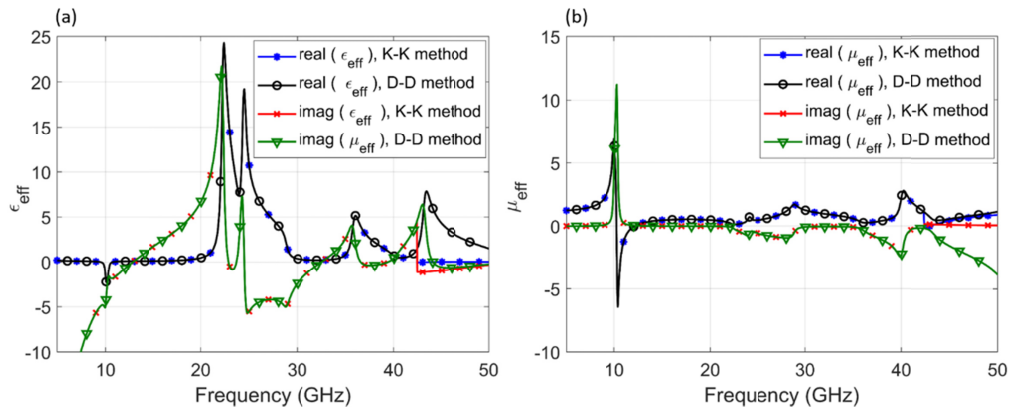


Fig. 18. Extracted effective (a) permittivity and (b) permeability of one-unit cell metamaterial, using the K-K method and the proposed method (D-D method).

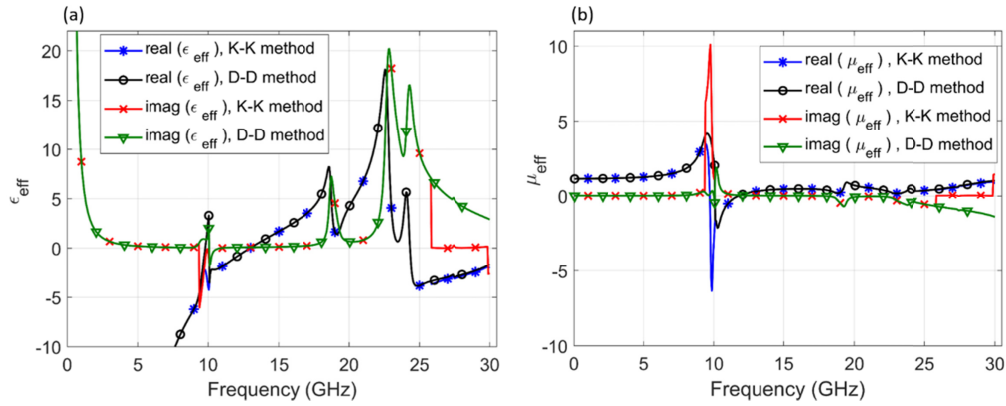


Fig. 19. Extracted effective (a) permittivity and (b) permeability of two-unit cell metamaterial, using the K-K method and the proposed method (D-D method).

6. Sensitivity to slab thickness and high frequency region

This section investigates the sensitivity of the proposed method and the K-K method at high frequencies for thick homogeneous metamaterial slabs. Figure 20 shows the behavior of the extracted n' using the K-K method and the proposed D-D method. The K-K method fails to extract correct values of n' beyond $\kappa d = 3.17$ even with a frequency range extension. In this investigation, $\kappa d = 3.17$ is the frequency saturation point of the K-K method. Figure 21 shows the n' extraction performance of the D-D method as compared to the K-K method. The proposed D-D method gives a linear (slope = 1) relationship between the frequency range of the correct results and the upper frequency limit used for extraction across the entire frequency axis. The extracted n' values across the entire frequency bands for the 200-nm and the 220-nm slabs are correct. For the K-K method, the frequency region $\kappa d < 3$ gives correct results with no need for frequency range extensions. Frequency range extension can help to enhance the performance of the K-K method to a specific saturation point, which is approximately $\kappa d = 3.17$, beyond which results are unreliable.

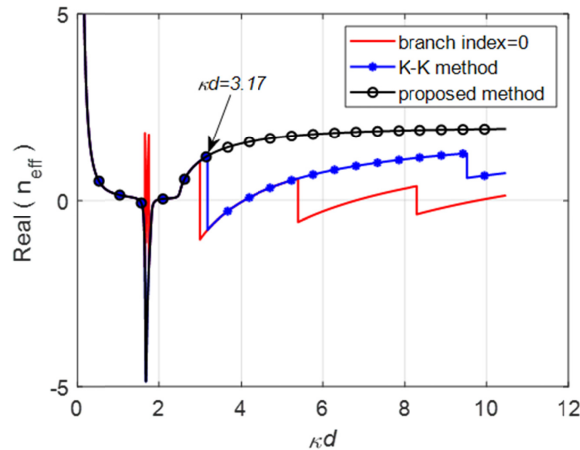


Fig. 20. Real refractive index of different frequency ranges and unit cell dimensions for K-K method and proposed method.

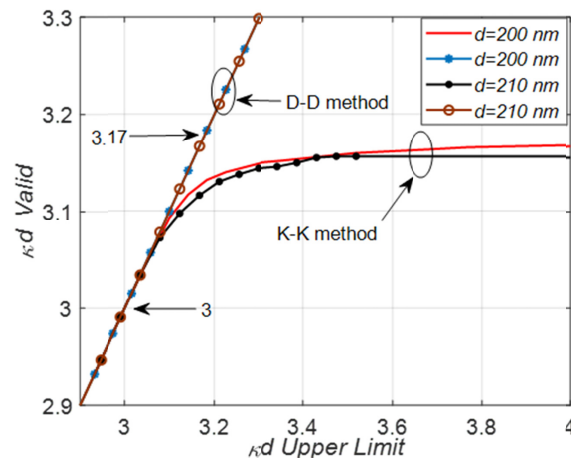


Fig. 21. Performance of proposed method and K–K method.

7. Conclusions

This paper presented an efficient retrieval method for extracting the effective metamaterial parameters using a DD discontinuity detection technique. This method is based on detecting the discontinuity points in the real part of the refractive index and estimating the correct branch value at these discontinuities. In contrast to the K–K method, the correct real part of n_{eff} is calculated independently from its imaginary part. The proposed method is analytically and numerically simple and does not involve an infinite frequency integration.

The performance of the proposed method was compared to the recently published results for the K–K method. Two cases were investigated in this study: uniform homogeneous slabs and SRR-rod metamaterial slabs. Both methods attained correct values of the branch index and, hence, the accurate values of n' at low frequency ranges for thin homogeneous and single-unit-cell SRR-rod slabs. However, the proposed method succeeded in overcoming the frequency and dimension limits observed in the K–K method in thick homogeneous and multi-cell SRR-rod slabs. The proposed method also gave notably accurate continuous values of the metamaterial parameters at resonance and at the negative parameter region.

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